

G. C. SCHOOL OF ENGINEERING AND COMPUTER SCIENCE
FINAL EXAMINATION FOR APPLIED DIFFERENTIAL EQUATIONS
ENGR 213 - FALL 2021

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Instructions:

Follow instructions of each problem and to provide solutions to all of them to receive full credit.

Only Faculty approved calculators are allowed

ALL PROBLEMS CARRY THE SAME WEIGHT

PROBLEM 1. Solve the initial value problem.

$$\frac{1}{x^2}y' - 3y^2 = 3; \quad y(1) = \frac{\pi}{4}$$

Solution:

$$\int \frac{dy}{1+y^2} = 3 \int x^2 dx$$

$$\arctan y = x^3 + C$$

$$C = 1 - \arctan\left(\frac{\pi}{4}\right) \approx -0.334$$

If variables are separated correctly: **3 points**

Integration: **4 points**

Constant found: **3 points**

PROBLEM 2. Find the implicit general solution of the following equation

$$(2xy\sin(xy) + x^2y^2\cos(xy) - 2xy)dx + (x^3y\cos(xy) + y\ln(y) + y)dy = 0$$

Solution:

$$\begin{cases} M(x,y) = 2xy\sin(xy) + x^2y^2\cos(xy) - 2xy \\ \Rightarrow M_y = 2x\sin(xy) + \underbrace{2x^2y\cos(xy) + 2x^2y\cos(xy) - x^3y^2\sin(xy) - 2x}_{\text{...}} \\ N(x,y) = x^3y\cos(xy) + y\ln(y) + y \\ \Rightarrow N_x = 3x^2y\cos(xy) - x^3y^2\sin(xy) \end{cases} \quad (2)$$

$$\frac{N_x - M_y}{M} = \frac{-x^2y\cos(xy) - 2x\sin(xy) + 2x}{2xy\sin(xy) + x^2y^2\cos(xy) - 2xy} = -\frac{1}{y}$$

integrating factor :

$$\mu = e^{\int \frac{N_x - M_y}{M} dy} = e^{-\int \frac{dy}{y}} = e^{-\ln y} = e^{\ln(1/y)} = 1/y \quad (2)$$

$$\begin{cases} \bar{M}(x,y) = 2x\sin(xy) + x^2y\cos(xy) - 2x \\ \bar{N}(x,y) = x^3\cos(xy) + \ln(y) + 1 \end{cases}$$

$$\begin{cases} \bar{M}_y = 2x^2\cos(xy) + x^2\cos(xy) - x^3y\sin(xy) = 3x^2\cos(xy) - x^3y\sin(xy) \\ \bar{N}_x = 3x^2\cos(xy) - x^3y\sin(xy) \end{cases}$$

$\Rightarrow$ Exact now ! (2)

$$\frac{\partial f}{\partial x} = \bar{M}(x, y) = 2x \sin(xy) + x^2 y \cos(xy) - 2x$$

$$\Rightarrow f(x, y) = 2 \int x \sin(xy) dx + y \int x^2 \cos(xy) dx - x^2 + g(y) = 2 \textcircled{I} + y \textcircled{II} - x^2 + g(y)$$

$$\textcircled{II} = \int \underbrace{x^2}_{u} \underbrace{\cos(xy)}_{dv} dx = \frac{x^2}{y} \sin(xy) - \frac{2}{y} \int x \sin(xy) dx$$

$u = x^2 \quad dv = \cos(xy)$
 $du = 2x dx \quad v = \frac{1}{y} \sin(xy)$

$u = x \quad dv = \sin(xy)$
 $du = dx \quad v = -\frac{1}{y} \cos(xy)$ \textcircled{I}

$$\Rightarrow \textcircled{II} = \frac{x^2}{y} \sin(xy) - \frac{2}{y} \left[-\frac{x}{y} \cos(xy) + \frac{1}{y} \int \cos(xy) dx \right]$$

$$= \frac{x^2}{y} \sin(xy) - \frac{2}{y} \left[-\frac{x}{y} \cos(xy) + \frac{1}{y^2} \sin(xy) \right] \quad \textcircled{I}$$

$$= \frac{x^2}{y} \sin(xy) + \frac{2x}{y^2} \cos(xy) - \frac{2}{y^3} \sin(xy)$$

$$\Rightarrow f(x, y) = -\cancel{\frac{2x}{y} \cos(xy)} + \cancel{\frac{2}{y^2} \sin(xy)} + \cancel{x^2 \sin(xy)} + \cancel{\frac{2x}{y} \cos(xy)} - \cancel{\frac{2}{y^2} \sin(xy)} - x^2 + g(y) \quad \textcircled{2}$$

$$\frac{\partial f}{\partial y} = x^3 \cos(xy) + g'(y) = \bar{N}(x, y) = x^3 \cos(xy) + \ln y + 1$$

$$\Rightarrow g'(y) = \ln y + 1 \Rightarrow g(y) = \int \ln y dy + y$$

$$\Rightarrow g(y) = y \ln(y) - y + y + c = y \ln(y) + c_1$$

$$\Rightarrow f(x, y) = x^2 \sin(xy) - x^2 + y \ln(y) + c_1 = c_2$$

$$\Rightarrow x^2 \sin(xy) - x^2 + y \ln(y) = c \quad \textcircled{2}$$

PROBLEM 3. A mass of 1 kg attached to a spring whose constant is 16 N/m stretches it and then comes to rest in the equilibrium position. Starting at $t = 0$, an external force equal to $f(t) = 4\cos(2t)$ is applied to the mass-spring system.

- (a) Find an equation of the motion if the surrounding medium offers a damping force equal to 8 times the instantaneous velocity.
 (b) Determine the transient term and the steady-state term in the equation of the motion.

Solution:

a) $m g = k s$; $m = 1$; $g = 9.81\text{ m/s}^2$; $k = 16 \Rightarrow s = \frac{9.8}{16} [\text{m}]$

$m X'' + \beta X' + k X = f(t)$; $m = 1, \beta = 8$; $k = 16$; $f(t) = 4 \cos(2t)$

$X'' + 8X' + 16X = 4 \cos(2t)$, $X(0) = 0$, $X'(0) = 0$ 2 pts

$m^2 + 8m + 16 = 0 \Rightarrow (m+4)^2 = 0$; $m_1 = m_2 = -4$

$X_c = c_1 e^{-4t} + c_2 t e^{-4t}$ 2 pts

$X_p = A \cos(2t) + B \sin(2t)$ ↗ $X_p' = -2A \sin(2t) + 2B \cos(2t)$

↘ $X_p'' = -4A \cos(2t) - 4B \sin(2t)$

$X_p'' + 8X_p' + 16X_p = -4A \cos(2t) - 4B \sin(2t) + 8[-2A \sin(2t) + 2B \cos(2t)] + 16[A \cos(2t) + B \sin(2t)]$

$= (2A + 16B) \cos(2t) + (12B - 16A) \sin(2t) = 4 \cos(2t) + 0 \sin(2t)$

$A = 3/25$; $B = 4/25$

$X(t) = c_1 e^{-4t} + c_2 t e^{-4t} + \frac{3}{25} \cos(2t) + \frac{4}{25} \sin(2t)$ 2 pts

use initial conditions :

$X(0) = 0 \Rightarrow c_1 + \frac{3}{25} = 0 \Rightarrow c_1 = -\frac{3}{25}$

$X'(0) = 0 \Rightarrow -4c_1 e^{-4t} + c_2 e^{-4t} - 4c_2 t e^{-4t} - \frac{6}{25} \sin(2t) + \frac{8}{25} \cos(2t) \Big|_{t=0} = 0$

$\frac{12}{25} + c_2 + \frac{8}{25} = 0 \Rightarrow c_2 = -\frac{4}{5}$ | $c_1 = -\frac{3}{25}$
 $t=0$

$X(t) = -\frac{1}{5} e^{-4t} \left(4t + \frac{3}{5} \right) + \frac{3}{25} \cos(2t) + \frac{4}{25} \sin(2t)$

2 pts

b) $-\frac{1}{5} e^{-4t} \left(4t + \frac{3}{5} \right)$ is the transient term of (a)

1 pt

because as $t \rightarrow \infty$ this term approaches zero

1 pt $\frac{3}{25} \cos(2t) + \frac{4}{25} \sin(2t)$ is the steady state term

PROBLEM 4. Solve the initial value problem below:

$$2y''' + 9y'' + 12y' + 5y = e^{-x}, y(0) = 0, y'(0) = 0, y''(0) = 0$$

Solution (Method of Variation of Parameters):

$$y = y_c + y_p$$

$$\boxed{y_c} \text{ Aux. Eqn. : } 2m^3 + 9m^2 + 12m + 5 = 0$$

$$(m+1)^2(2m+5) = 0$$

$$m_{1,2} = -1 \quad m_3 = -5/2 = -2.5$$

$$\Rightarrow \boxed{y_c = c_1 e^{-x} + c_2 x e^{-x} + c_3 e^{-2.5x}}$$

(1)

$$\boxed{y_p} \text{ Variation of parameters : } y_p = u_1 y_1 + u_2 y_2 + u_3 y_3$$

$$\text{Standard form : } y'' + 4.5y' + 6y = \frac{1}{2}e^{-x}$$

(0.5)

$$\boxed{W} = \begin{vmatrix} e^{-x} & x e^{-x} & e^{-2.5x} \\ -e^{-x} & e^{-x} - x e^{-x} & -2.5 e^{-2.5x} \\ e^{-x} & -2e^{-x} + x e^{-x} & 6.25 e^{-2.5x} \end{vmatrix} = \begin{vmatrix} e^{-x} & x e^{-x} & e^{-2.5x} \\ 0 & e^{-x} & -1.5 e^{-2.5x} \\ 0 & -2e^{-x} & 5.25 e^{-2.5x} \end{vmatrix}$$

$$= e^{-x} [5.25 - 3] e^{-3.5x} = 2.25 e^{-4.5x}$$

(1)

$$\boxed{W_1} = \begin{vmatrix} 0 & x e^{-x} & e^{-2.5x} \\ 0 & e^{-x} - x e^{-x} & -2.5 e^{-2.5x} \\ \frac{1}{2} e^{-x} & -2e^{-x} + x e^{-x} & 6.25 e^{-2.5x} \end{vmatrix}$$

$$= \frac{1}{2} e^{-x} [-2.5x e^{-3.5x} - e^{-3.5x} (1-x)] = -e^{-4.5x} (0.75x + 0.5)$$

(1)

$$W_2 = \begin{vmatrix} e^{-x} & 0 & e^{-2.5x} \\ -e^{-x} & 0 & -2.5e^{-2.5x} \\ e^{-x} & \frac{1}{2}e^{-x} & 6.25e^{-2.5x} \end{vmatrix}$$

$$= -\frac{1}{2}e^{-x} \left[-2.5e^{-3.5x} + e^{-3.5x} \right] = 0.75e^{-4.5x} \quad (1)$$

$$W_3 = \begin{vmatrix} e^{-x} & xe^{-x} & 0 \\ -e^{-x} & e^{-x} - xe^{-x} & 0 \\ e^{-x} & -2e^{-x} + xe^{-x} & \frac{1}{2}e^{-x} \end{vmatrix}$$

$$= \frac{1}{2}e^{-x} \left[e^{-2x}(1-x) + xe^{-2x} \right] = 0.5e^{-3x} \quad (1)$$

$$u'_1 = \frac{W_1}{W} = -\frac{1}{3}x - \frac{2}{9} \Rightarrow u_1 = -\frac{1}{6}x^2 - \frac{2}{9}x \quad (0.5)$$

$$u'_2 = \frac{W_2}{W} = \frac{1}{3} \Rightarrow u_2 = \frac{1}{3}x \quad (0.5)$$

$$u'_3 = \frac{W_3}{W} = \frac{2}{9}e^{3/2x} \Rightarrow u_3 = \frac{4}{27}e^{3x/2} \quad (0.5)$$

$$y_p = -\frac{1}{6}x^2e^{-x} - \frac{2}{9}xe^{-x} + \frac{1}{3}x^2e^{-x} + \frac{4}{27}e^{-x} = \frac{1}{6}x^2e^{-x} - \frac{2}{9}xe^{-x} + \frac{4}{27}e^{-x} \quad (0.5)$$

$$\Rightarrow y = y_c + y_p = e^{-x} \left[\underbrace{(c_1 + \frac{4}{27})}_{c_4} + \underbrace{(c_2 - \frac{2}{9})x}_{c_5} + \frac{1}{6}x^2 + c_3e^{-1.5x} \right]$$

$$\Rightarrow y = c_4e^{-x} + c_5xe^{-x} + c_3e^{-2.5x} + \frac{1}{6}x^2e^{-x} \quad (0.5)$$

$$y' = -c_4e^{-x} + c_5e^{-x} - c_5xe^{-x} - 2.5c_3e^{-2.5x} + \frac{1}{3}xe^{-x} - \frac{1}{6}x^2e^{-x}$$

$$y'(0) = 0 \Rightarrow -c_4 + c_5 - 2.5c_3 = 0 \Rightarrow c_5 = 1.5c_3$$

$$y(0) = 0 \Rightarrow c_4 + c_3 = 0 \Rightarrow c_4 = -c_3$$

$$y'' = c_4e^{-x} - c_5e^{-x} - c_5e^{-x} + c_5xe^{-x} + 6.25c_3e^{-2.5x} + \frac{1}{3}e^{-x} - \frac{1}{3}xe^{-x} - \frac{1}{3}xe^{-x} + \frac{1}{6}x^2e^{-x}$$

$$y''(0) = c_4 - 2c_5 + 6.25c_3 + \frac{1}{3} = 0 \Rightarrow 2.25c_3 = -\frac{1}{3} \Rightarrow c_3 = -0.148 = -\frac{4}{27}$$

$$\Rightarrow y = 0.148e^{-x} - 0.222xe^{-x} - 0.148e^{-2.5x} + \frac{1}{6}x^2e^{-x}$$

$$\begin{matrix} \swarrow & \downarrow & \searrow \\ \frac{4}{27} & -\frac{2}{9} & -\frac{4}{27} \end{matrix}$$

(2)

Alternative Solution (Method of undetermined Coefficients):

$$y_c = c_1 e^{-x} + c_2 x e^{-x} + c_3 e^{-2.5x}$$

(2)

y_p Undetermined Coefficients: $y_p = A e^{-x}$

But repeated in y_c then $\rightarrow y_p = A x^2 e^{-x}$

(1)

(1) $y_p' = 2Ax e^{-x} - A x^2 e^{-x}$

(1) $y_p'' = 2A e^{-x} - 2Ax e^{-x} - 2Ax e^{-x} + A x^2 e^{-x} = 2A e^{-x} - 4Ax e^{-x} + A x^2 e^{-x}$

(1) $y_p''' = -2A e^{-x} - 4A e^{-x} + 4Ax e^{-x} + 2Ax e^{-x} - A x^2 e^{-x}$
 $= -6A e^{-x} + 6Ax e^{-x} - A x^2 e^{-x}$

$$2y''' + 9y'' + 12y' + 5y = -12A e^{-x} + 12Ax e^{-x} - 2Ax^2 e^{-x} + 18A e^{-x} - 36Ax e^{-x} + 9Ax^2 e^{-x} + 24Ax e^{-x} - 12Ax^2 e^{-x} + 5Ax^2 e^{-x} = e^{-x}$$

(1) $\Rightarrow -12A + 18A = 1 \Rightarrow A = \frac{1}{6}$

(1) $\Rightarrow y = y_c + y_p = c_1 e^{-x} + c_2 x e^{-x} + c_3 e^{-2.5x} + \frac{1}{6} x^2 e^{-x}$

$$y'(0) = -c_1 + c_2 - 2.5c_3 = 0 \Rightarrow c_1 - c_2 + 2.5c_3 = 0 \Rightarrow c_2 = -1.5c_1$$

$$y(0) = c_1 + c_3 = 0 \Rightarrow c_3 = -c_1$$

$$y''(0) = c_1 - 2c_2 + 6.25c_3 + \frac{1}{3} = 0 \Rightarrow c_1 = 0.148^{4/27}, c_2 = -0.222^{2/4}, c_3 = -0.148^{4/27}$$

(2) $\Rightarrow y = 0.148 e^{-x} - 0.222 x e^{-x} - 0.148 e^{-2.5x} + \frac{1}{6} x^2 e^{-x}$

PROBLEM 5. Use Bernoulli's method to solve the initial value differential equation and express your solution in explicit format.

$$x^2 y' + 2xy - y^3 = 0 \quad y(1) = 2$$

Solution:

$$x^2 y' + 2xy - y^3 = 0 \quad (1)$$

$$x^2 y' + 2xy = y^3$$

$$y' + (2/x)y = (1/x^2)y^3 \quad \text{divide by } y^3$$

$$y^{-3} y' + (2/x)y^{-2} = 1/x^2$$

$$\text{Let } u = y^{-2} \quad u' = -2y^{-3} y' \Rightarrow y' = -\frac{1}{2} y^3 u' \quad (2)$$

Substitute back in the given equation; $(-1/2)u' + (2/x)u = 1/x^2$

$$u' - (4/x)u = -2/x^2$$

$$\text{integrating factor } e^{\int \frac{-4}{x} dx} = e^{-4 \ln(x)} = e^{\ln(\frac{1}{x})^4} = 1/x^4 \quad (2)$$

$$(1/x)^4 u = \int \frac{1-2}{x^4 x^2} dx = \int \frac{-2}{x^6} dx = \frac{2/5}{x^5} + c$$

$$u = (2/5x) + cx^4$$

$$y^{-2} = (2/5x) + cx^4 = \frac{2 + 5cx^5}{5x} \quad (2)$$

$$y^2 = \frac{5x}{2 + 5cx^5} \quad \text{apply the I.C.} \quad 4 = \frac{5}{2 + 5c} \Rightarrow 5c = -0.75 \quad (2)$$

$$y^2 = \frac{5x}{2 - 0.75x^5}$$

$$y = \sqrt{\frac{5x}{2 - 0.75x^5}} \quad (1)$$

PROBLEM 6. Solve the following differential equation with the method of your choice.

$$x^2 y''' = 2y'$$

Solution:

$$z = y'$$

$$x^2 z'' - 2z = 0$$

$$\alpha(\alpha - 1) - 2 = 0$$

$$\alpha = -1, 2$$

$$z = C_1 \frac{1}{x} + C_2 x^2$$

$$y = C_1 \ln x + C_2 x^3 + C_3$$

Change of variable $z = y'$ made: **2 points**

Characteristic equation solved correctly : **3 points**

Solution of Cauchy-Euler D. E. of order 2 is correctly written: **2 points**

Integration $y = \int z$ done: **3 points**

PROBLEM 7. Find the general solution of the following differential equation:



$$4y'' + 36y' = \csc(3x)$$

Solution:

Put the equation in normal form and create the auxiliary equation:

$$y'' + 9y = \frac{1}{4} \csc(3x) \quad m^2 + 9 = 0$$



Solving the auxiliary equation, we find $m = 3i, -3i$ and so we get the solutions to the homogeneous equation is

$$y_c = c_1 \cos(3x) + c_2 \sin(3x)$$

Since $\csc$ is not a nice function we use variation of parameters to get that the particular solution is of the form

$$y_p = u_1(x) \cos(3x) + u_2(x) \sin(3x)$$

Applying the Wronskian theory, we find

$$W = \det \begin{pmatrix} \cos(3x) & \sin(3x) \\ -3 \sin(3x) & 3 \cos(3x) \end{pmatrix} = 3$$

$$W_1 = \det \begin{pmatrix} 0 & \sin(3x) \\ \frac{1}{4} \csc(3x) & 3 \cos(3x) \end{pmatrix} = -\frac{1}{4}$$

$$W_2 = \det \begin{pmatrix} \cos(3x) & 0 \\ -3 \sin(3x) & \frac{1}{4} \csc(3x) \end{pmatrix} = \frac{1}{4} \frac{\cos(3x)}{\sin(3x)}$$

Hence

$$u_1' = \frac{W_1}{W} = -\frac{1}{12} \Rightarrow u_1 = -\frac{1}{12}x$$

$$u_2' = \frac{W_2}{W} = \frac{1}{12} \frac{\cos(3x)}{\sin(3x)} \Rightarrow u_2 = \frac{1}{36} \ln |\sin(3x)|$$

Therefore, a particular solution to the ODE would be

$$y_p = -\frac{1}{12}x \cos(3x) + \frac{1}{36} \sin(3x) \ln |\sin(3x)|$$

and the general solution would be

$$y = c_1 \cos(3x) + c_2 \sin(3x) - \frac{1}{12}x \cos(3x) + \frac{1}{36} \sin(3x) \ln |\sin(3x)|$$

1. 1 point for finding and solving auxiliary equation
2. 2 point for finding $\cos(3x)$ and $\sin(3x)$ and writing y_c
3. 3 points for computing the 3 correct Wronskians
4. 1 point for determining u_1
5. 2 points for determine u_2
6. 1 point for writing the general solution

PROBLEM 8. Two chemicals A and B are combined to form a chemical C. The rate of the reaction is proportional to the product of instantaneous amounts of A and B not converted to C. Initially there are 40g of A and 30g of B, and for 3g of B 2g of A are used. It is observed that 5g of C is formed in 2 minutes. How much C is formed after time t. What is the limiting amount of C as $t \rightarrow \infty$?

Solution:

$$\frac{dx}{dt} = k_1 \left(a - \frac{M}{M+N} x \right) \left(b - \frac{N}{M+N} x \right)$$

a: initial amount of A

b: initial amount of B

M: parts of A that should be used in making C

N: parts of B that should be used in making C

$$\frac{dx}{dt} = k_1 \left(40 - \frac{2}{2+3} x \right) \left(30 - \frac{3}{2+3} x \right) = k_2 (200 - 2x)(150 - 3x)$$

marks
3

$$\frac{dx}{dt} = K (100 - x)(50 - x)$$

$$K_2 = \frac{K_1}{25}, \quad K = 6 K_2 = \frac{6K_1}{25}$$

$$\Rightarrow \int \frac{dx}{(100-x)(50-x)} = Kt + C_1 \quad (I)$$

$$\frac{1}{(100-x)(50-x)} = \frac{A}{100-x} + \frac{B}{50-x} \Rightarrow \begin{cases} 50A + 100B = 1 \\ -A - B = 0 \Rightarrow B = -A \\ -50A = 1 \Rightarrow A = -\frac{1}{50} \\ B = \frac{1}{50} \end{cases}$$

$$(I) \Rightarrow \int \frac{-1/50}{100-x} dx + \int \frac{1/50}{50-x} = Kt + C_1$$

$$\frac{1}{50} \ln |100-x| - \frac{1}{50} \ln |50-x| = Kt + C_1$$

$$\ln \left| \frac{50-x}{100-x} \right| = -50Kt + C_2$$

$$C_2 = -50C_1$$

marks
3

$$\Rightarrow \frac{50-x}{100-x} = C e^{-50Kt}$$

$$C = e^{C_2}$$

$$\text{at } t=0 \quad x=0 \Rightarrow \frac{1}{2} = C e^0 \Rightarrow C = \frac{1}{2}$$

marks
2

$$\text{at } t=2 \text{ min } x=5 \text{ gr} \Rightarrow \frac{45}{95} = \frac{1}{2} \cdot e^{-100K} \Rightarrow \frac{1}{2} \ln \left(\frac{18}{19} \right) = -50K$$

$$\Rightarrow -50K = -0.027$$

marks
1

$$\Rightarrow \frac{50-x}{100-x} = \frac{1}{2} e^{-0.027t}$$

$$\text{or } x = 50 \frac{1 - e^{-0.027t}}{1 - 0.5 e^{-0.027t}}$$

marks
1

$$\text{When } t \rightarrow \infty \quad 50-x=0 \Rightarrow x=50 \text{ gr}$$

PROBLEM 9. Find the general solution the following system of differential equations.

$$X' = \begin{pmatrix} 1 & 8 \\ 1 & -1 \end{pmatrix} X + \begin{pmatrix} e^{-t} \\ 6 \end{pmatrix}$$

Solution:

$$\begin{aligned} \text{Det}(A-\lambda I) &= \begin{vmatrix} 1-\lambda & 8 \\ 1 & -1-\lambda \end{vmatrix} = (1-\lambda)(-1-\lambda)-8=0 \\ -1+\lambda^2-8 &= 0 & \lambda^2-9=0 & \lambda=3 \text{ Or } -3 \end{aligned} \quad (2)$$

For $\lambda=3$

$$\begin{pmatrix} -2 & 8 \\ 1 & -4 \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = 0; k_1 - 4k_2 = 0; k_1 = 4k_2; K_1 = \begin{pmatrix} 4 \\ 1 \end{pmatrix} \quad (1)$$

For $\lambda=-3$

$$\begin{pmatrix} 4 & 8 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} = 0; k_1 + 2k_2 = 0; k_1 = -2k_2; K_2 = \begin{pmatrix} -2 \\ 1 \end{pmatrix} \quad (1)$$

$$X_c = c_1 \begin{pmatrix} 4 \\ 1 \end{pmatrix} e^{3t} + c_2 \begin{pmatrix} -2 \\ 1 \end{pmatrix} e^{-3t}$$

Using Variation of parameter method

$$\Phi(t) = \begin{pmatrix} 4e^{3t} & -2e^{-3t} \\ e^{3t} & e^{-3t} \end{pmatrix} \quad \det \Phi(t) = 4+2=6 \quad \Phi(t)^{-1} = \begin{pmatrix} 1/6 e^{-3t} & 1/3 e^{-3t} \\ -1/6 e^{3t} & 2/3 e^{3t} \end{pmatrix} \quad (2)$$

$$\Phi(t)^{-1} F(t) = \begin{pmatrix} 1/6 e^{-3t} & 1/3 e^{-3t} \\ -1/6 e^{3t} & 2/3 e^{3t} \end{pmatrix} \begin{pmatrix} e^{-t} \\ 6 \end{pmatrix} = \begin{pmatrix} 1/6 e^{-4t} + 2e^{-3t} \\ -1/6 e^{2t} + 4e^{3t} \end{pmatrix}$$

$$\int \Phi(t)^{-1} F(t) dt = \int \begin{pmatrix} 1/6 e^{-4t} + 2e^{-3t} \\ -1/6 e^{2t} + 4e^{3t} \end{pmatrix} dt = \begin{pmatrix} -1/24 e^{-4t} - 2/3 e^{-3t} \\ -1/12 e^{2t} + 4/3 e^{3t} \end{pmatrix} \quad (2)$$

$$X_p = \Phi(t) \int \Phi(t)^{-1} F(t) dt = \begin{pmatrix} 4e^{3t} & -2e^{-3t} \\ e^{3t} & e^{-3t} \end{pmatrix} \begin{pmatrix} -1/24 e^{-4t} - 2/3 e^{-3t} \\ -1/12 e^{2t} + 4/3 e^{3t} \end{pmatrix} = \begin{pmatrix} -16/3 \\ -1/8 e^{-t} + 2/3 \end{pmatrix} \quad (1)$$

$$X = X_c + X_p = c_1 \begin{pmatrix} 4 \\ 1 \end{pmatrix} e^{3t} + c_2 \begin{pmatrix} -2 \\ 1 \end{pmatrix} e^{-3t} + \begin{pmatrix} -16/3 \\ -1/8 e^{-t} + 2/3 \end{pmatrix} = c_1 \begin{pmatrix} 4 \\ 1 \end{pmatrix} e^{3t} + c_2 \begin{pmatrix} -2 \\ 1 \end{pmatrix} e^{-3t} + \begin{pmatrix} 0 \\ -1/8 \end{pmatrix} e^{-t} + \begin{pmatrix} -16/3 \\ 2/3 \end{pmatrix} \quad (1)$$

Using undetermined coefficients

$$F(t) = \begin{pmatrix} e^{-t} \\ 6 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{-t} + \begin{pmatrix} 0 \\ 6 \end{pmatrix}$$

$$X_p = \begin{pmatrix} A \\ B \end{pmatrix} e^{-t} + \begin{pmatrix} C \\ D \end{pmatrix} \quad X'_p = -\begin{pmatrix} A \\ B \end{pmatrix} e^{-t} \quad (2)$$

$$-\begin{pmatrix} A \\ B \end{pmatrix} e^{-t} = \begin{pmatrix} 1 & 8 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} A \\ B \end{pmatrix} e^{-t} + \begin{pmatrix} 1 & 8 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} C \\ D \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{-t} + \begin{pmatrix} 0 \\ 6 \end{pmatrix}$$

$$-\begin{pmatrix} A \\ B \end{pmatrix} e^{-t} = \begin{pmatrix} A+8B \\ A-B \end{pmatrix} e^{-t} + \begin{pmatrix} C+8D \\ C-D \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} e^{-t} + \begin{pmatrix} 0 \\ 6 \end{pmatrix} = \begin{pmatrix} A+8B+1 \\ A-B \end{pmatrix} e^{-t} + \begin{pmatrix} C+8D \\ C-D+6 \end{pmatrix}$$

$$-A=A+8B+1; -B=A-B; C+8D=0; C-D+6=0 \text{ Solve that give } A=0; B=-1/8; C=-16/3; D=2/3$$

$$X_p = \begin{pmatrix} 0 \\ -1/8 \end{pmatrix} e^{-t} + \begin{pmatrix} -16/3 \\ 2/3 \end{pmatrix} \quad (3)$$

$$X = X_c + X_p = c_1 \begin{pmatrix} 4 \\ 1 \end{pmatrix} e^{3t} + c_2 \begin{pmatrix} -2 \\ 1 \end{pmatrix} e^{-3t} + \begin{pmatrix} -16/3 \\ -1/8 e^{-t} + 2/3 \end{pmatrix} = c_1 \begin{pmatrix} 4 \\ 1 \end{pmatrix} e^{3t} + c_2 \begin{pmatrix} -2 \\ 1 \end{pmatrix} e^{-3t} + \begin{pmatrix} 0 \\ -1/8 \end{pmatrix} e^{-t} + \begin{pmatrix} -16/3 \\ 2/3 \end{pmatrix} \quad (1)$$

PROBLEM 10. Find the first six terms of the power-series solution of the following linear ODE.

$$(x-1)y'' - x^2y' + xy = 0$$

Solution:

$$(x-1)y'' - x^2y' + xy = 0 \quad \textcircled{I}$$

Answer:

$$y = \sum_{n=0}^{\infty} c_n x^n \Rightarrow y' = \sum_{n=1}^{\infty} n c_n x^{n-1}, \quad y'' = \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2}$$

$$\textcircled{I} \Rightarrow (x-1) \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} - x^2 \sum_{n=1}^{\infty} n c_n x^{n-1} + x \sum_{n=0}^{\infty} c_n x^n = 0 \quad \textcircled{1}$$

$$\Rightarrow 2c_2 x - 2c_2 - 6c_3 x + c_0 x + \sum_{n=3}^{\infty} n(n-1) c_n x^{n-1} - \sum_{n=4}^{\infty} n(n-1) c_n x^{n-2} - \sum_{n=1}^{\infty} n c_n x^{n+1} + \sum_{n=1}^{\infty} c_n x^{n+1} = 0 \quad \textcircled{1}$$

$$\Rightarrow -2c_2 + (c_0 + 2c_2 - 6c_3)x + \sum_{k=2}^{\infty} k(k+1) c_{k+1} x^k - \sum_{k=2}^{\infty} (k+1)(k+2) c_{k+2} x^k - \sum_{k=2}^{\infty} (k-2) c_{k-1} x^k = 0 \quad \textcircled{1}$$

$$\begin{aligned} n-1=k &\Rightarrow n=k+1 \\ n-2=k &\Rightarrow n=k+2 \\ n+1=k &\Rightarrow n=k-1 \end{aligned}$$

$$\Rightarrow -2c_2 + (c_0 + 2c_2 - 6c_3)x + \sum_{k=2}^{\infty} [k(k+1) c_{k+1} - (k+1)(k+2) c_{k+2} - (k-2) c_{k-1}] x^k = 0$$

$$\Rightarrow \boxed{c_2 = 0}, \quad c_0 - 6c_3 = 0 \Rightarrow \boxed{c_3 = \frac{1}{6} c_0} \quad \textcircled{1}$$

$$k(k+1) c_{k+1} - (k+1)(k+2) c_{k+2} - (k-2) c_{k-1} = 0 \Rightarrow \boxed{c_{k+2} = \frac{k(k+1) c_{k+1} - (k-2) c_{k-1}}{(k+1)(k+2)}}$$

Three-term recurrence relation, then :

$$k = 2, 3, 4, \dots$$

① $c_0 \neq 0, c_1 = 0 \Rightarrow y_1$ in terms of c_0

② $c_0 = 0, c_1 \neq 0 \Rightarrow y_2$ in terms of c_1

and then $y = c_0 y_1 + c_1 y_2$

$$C_0 \neq 0, C_1 = 0$$

$$C_2 = 0$$

$$C_3 = \frac{1}{6} C_0$$

$$K=2 \Rightarrow C_4 = \frac{6C_3 - 0C_1}{12} = \frac{1}{2} C_3 = +\frac{1}{12} C_0$$

$$K=3 \Rightarrow C_5 = \frac{12C_4 - 3C_2}{20} = \frac{3}{5} C_4 = +\frac{1}{20} C_0$$

$$K=4 \Rightarrow C_6 = \frac{20C_5 - 2C_3}{30} = \frac{C_0 - \frac{1}{3}C_0}{30} = \frac{1}{45} C_0$$

$$\Rightarrow C_0 y_1 = \sum_{n=0}^{\infty} C_n x^n = C_0 + \frac{1}{6} C_0 x^3 + \frac{1}{12} C_0 x^4 + \frac{1}{20} C_0 x^5 + \frac{1}{45} C_0 x^6 + \dots$$

$$\Rightarrow y_1 = 1 + \frac{1}{6} x^3 + \frac{1}{12} x^4 + \frac{1}{20} x^5 + \frac{1}{45} x^6 + \dots \quad (3)$$

$$C_0 = 0, C_1 \neq 0$$

$$C_2 = 0$$

$$C_3 = \frac{1}{6} C_0 = 0$$

$$C_4 = \frac{6C_3 - 0C_1}{12} = 0$$

$$C_5 = \frac{12C_4 - 0C_2}{20} = \frac{3}{5} C_4 = 0$$

$$C_6 = \frac{20C_5 - 2C_3}{30} = 0$$

$$\Rightarrow C_1 y_2 = \sum_{n=0}^{\infty} C_n x^n = C_1 x$$

$$\Rightarrow y_2 = x \quad (3)$$

$$\text{Then } y = C_0 y_1 + C_1 y_2 = C_0 \left(1 + \frac{1}{6} x^3 + \frac{1}{12} x^4 + \frac{1}{20} x^5 + \frac{1}{45} x^6 + \dots \right) + C_1 x$$